G-ideals, Hilbert rings, and the Nullstellensatz

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A modern style: the first thing to do with a maximal ideal $\mathfrak{m}$ in $R[x]$ is to contract it back to the ideal $\mathfrak{m} \cap R$ in R where R is more or less aebitrary commutative ring. When we do this we certainly get a prime ideal in R, but we wish it were a maximal ideal. A close examination of the facts leads to the sequence of ideals: *G-domain, G-ideal, and Hilbert ring*.

Let R be an integral domian with quotient field K. The following two statements are equivalent: 1. K is a finitely generated R-algebra. 2. as a ring, K can be generated over R by one element.